

AI Career Intelligence Platform

Operations Design Document

Monitoring • Logging • Backup • Disaster Recovery

Version: 1.0

Prepared By: Akshay Mohan

Table of Contents

- 1. Monitoring
- 2. Logging
- 3. Backup Strategy
- 4. Disaster Recovery
- 5. Conclusion

1. Monitoring

Continuous monitoring helps ensure the platform remains available, reliable, and performant.

Metric	Description
CPU & Memory	Monitor server resource utilization
API Response Time	Track request latency
Database Health	Monitor PostgreSQL performance
Redis Health	Monitor cache availability
Application Uptime	Track service availability

Recommended Tools: Prometheus, Grafana, Uptime Kuma.

2. Logging

Application and infrastructure logs help diagnose issues and support auditing.

Log Type	Purpose
Application Logs	Business logic and errors
API Logs	Request and response tracking
Security Logs	Authentication and access events
System Logs	Operating system and infrastructure events
Audit Logs	Administrative actions

Recommended Stack: ELK (Elasticsearch, Logstash, Kibana) or Loki + Grafana.

3. Backup Strategy

Asset	Backup Frequency
PostgreSQL Database	Daily
Uploaded Resumes	Daily
Configuration Files	Weekly
Docker Images	Per Release
Application Source Code	Git Repository

- Store backups in a secure off-site location.
- Encrypt backup files.
- Periodically verify backup restoration.

4. Disaster Recovery

The disaster recovery plan minimizes downtime and ensures business continuity.

Recovery Activity	Description
Database Restore	Recover latest verified backup
Application Restore	Redeploy containers
Configuration Restore	Recover infrastructure settings
Health Verification	Validate all services
Rollback	Return to previous stable release if required

Objectives: RPO $\leq$ 24 hours, RTO $\leq$ 4 hours (target values for Version 1.0).

5. Conclusion

This document defines the operational practices for Version 1.0 of the AI Career Intelligence Platform, covering monitoring, centralized logging, backup management, and disaster recovery planning to ensure reliable production operations.